TUM Institute for Cognitive Systems (ICS)
Multi-Sensory Based Robot Dynamic Manipulation

Tutorial: How to use the Schunk WSG-50 gripper

Dr.-Ing. Rogelio Guadarrama

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This tutorial explains how to start the gripper driver and command it to open and close the gripper.

IMPORTANT: If you notice anything strange with the gripper or its cable, do not operate the robot and contact one of the Course supervisors.

The gripper is very strong. Do not put your hand between the fingers.

The Schunk WSG-50 gripper mounted on the robot is very simple to use. But you must launch the bringup node with the whole kinematic chain when you launch the UR10 robot. Follow these steps:

1. Start and initialize the UR10 robot as shown in the last tutorial.

2. Verify if you can ping the gripper (IP: 192.168.1.20).

3. Run the UR script manager.

   ```
   roslaunch tum_ics_ur_robot_manager robot_script_manager_ur10.launch
   ```

4. Launch the WSG-50 gripper driver.

   ```
   roslaunch wsg_50_driver wsg_50_tcp.launch
   ```

5. Launch the rviz visualization.

   ```
   roslaunch tum_ics_ur10_bringup bringUR10-FT-wsg-50.launch
   ```

6. Launch your controller.

   ```
   roslaunch tum_ics_ur10_controller_tutorial testSimpleEffortCtrl.launch
   ```

7. The gripper driver can be commanded with different ROS services. The services you need are:

```
rosservice call /wsg_50_driver/grasp "width 0.0
speed:  50.0"
```

and

```
rosservice call /wsg_50_driver/release "width 88.0
speed:  50.0"
```

width is the desired separation between the fingers measured in **mm**. The maximum possible width is 88 mm. The minimum is 0 mm.

"speed" is the finger displacement speed measured in mm/s. Do not use higher speeds than 100 mm/s.